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COURSE: AGILE DEVELOPMET LAB

FAT EAXM

INVENTORY LEVELS ARE UPDATAED ACCURATELY AFTER STOCK IMAGES:

Deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: java-app
  template:
    metadata:
      labels:
        app: java-app
    spec:
      containers:
      - name: java-container
        image: 23mis0407/java-app:latest
        ports:
        - containerPort: 8080
---
```

```
apiVersion: v1
kind: Service
metadata:
  name: java-service
spec:
  type: NodePort
  selector:
    app: java-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30007
```

Dockerfile:

```
# Use stable Java 17 image
FROM eclipse-temurin:17-jdk-jammy

WORKDIR /app

COPY target/DISPLAYRESULT_MAVEN-1.0-SNAPSHOT.jar
app.jar

ENTRYPOINT ["java", "-jar", "app.jar"]
```

Inventory.java:

```java
package com.inventory;

import java.util.HashMap;
import java.util.Map;

public class InventoryService {

    private Map<String, Integer> inventory = new HashMap<>();

    // Add stock
    public void addStock(String item, int quantity) {
        inventory.put(item, inventory.getOrDefault(item, 0) + quantity);
    }

    // Remove stock
    public void removeStock(String item, int quantity) {
        if (inventory.containsKey(item)) {
            int current = inventory.get(item);
            if (current >= quantity) {
                inventory.put(item, current - quantity);
            } else {
                throw new IllegalArgumentException("Not enough stock");
            }
        }
```

```
    }

    // Get stock
    public int getStock(String item) {
        return inventory.getOrDefault(item, 0);
    }
}
```

Junittest.java:

```
package com.inventory;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class InventoryServiceTest {

    @Test
    public void testAddStock() {
        InventoryService service = new InventoryService();
        service.addStock("Pen", 10);
        assertEquals(10, service.getStock("Pen"));
    }

    @Test
    public void testRemoveStock() {
```

```
        InventoryService service = new InventoryService();
        service.addStock("Pen", 10);
        service.removeStock("Pen", 5);
        assertEquals(5, service.getStock("Pen"));
    }

    @Test
    public void testRemoveMoreThanAvailable() {
        InventoryService service = new InventoryService();
        service.addStock("Pen", 5);

        assertThrows(IllegalArgumentException.class, () -> {
            service.removeStock("Pen", 10);
        });
    }
}
```

Pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>DISPLAYRESULT_MAVEN</artifactId>
```

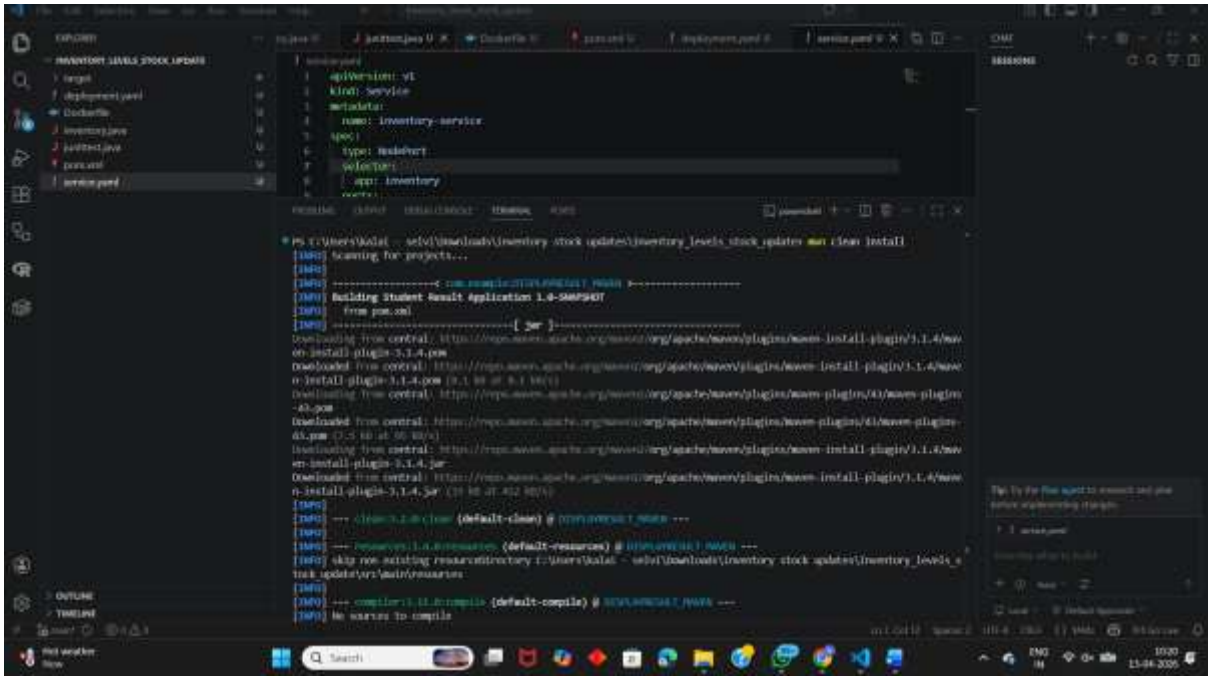
```
<version>1.0-SNAPSHOT</version>
<name>Student Result Application</name>
<properties>
<maven.compiler.source>17</maven.compiler.source>
<maven.compiler.target>17</maven.compiler.target>
<project.build.sourceEncoding>UTF-
8</project.build.sourceEncoding>
</properties>
<dependencies>
<dependency>
<groupId>org.junit.jupiter</groupId>
<artifactId>junit-jupiter</artifactId>
<version>5.10.0</version>
<scope>test</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
</plugin>
<plugin>
```

```
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>3.1.2</version>
</plugin>
</plugins>
</build>
</project>
```

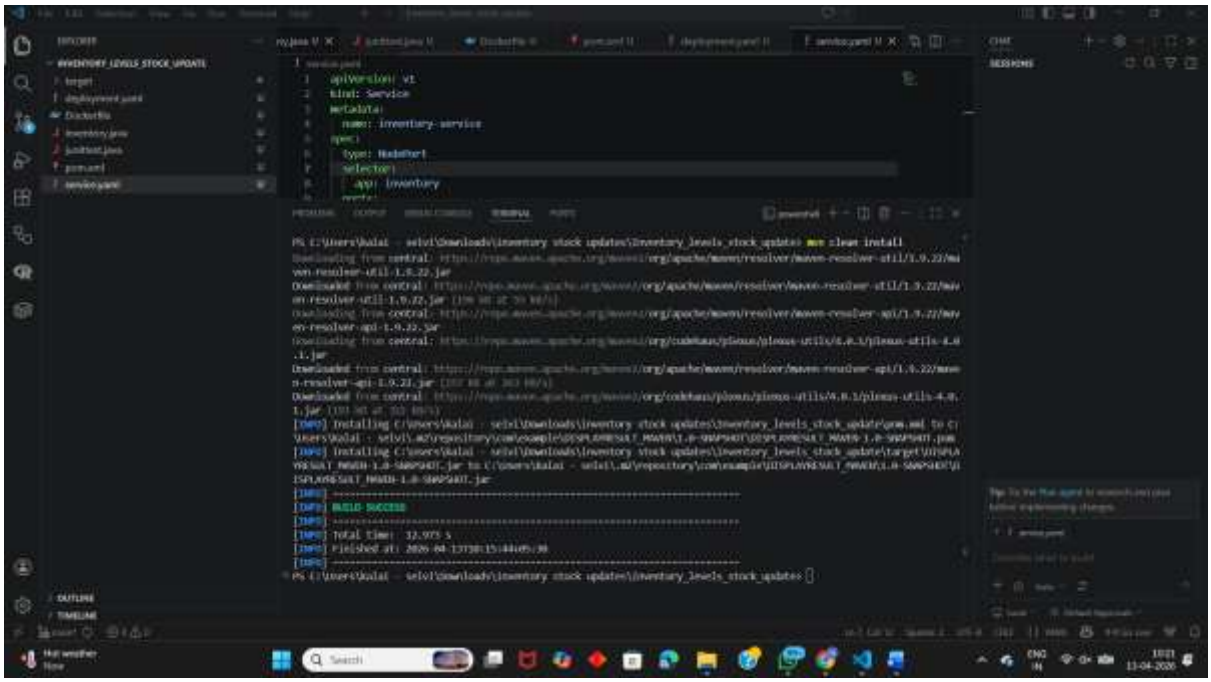
Service.yaml:

```
apiVersion: v1
kind: Service
metadata:
  name: inventory-service
spec:
  type: NodePort
  selector:
    app: inventory
  ports:
    - port: 80
      targetPort: 8080
```

MAVEN RUNNING:



.



.

Push into the github:

Git init

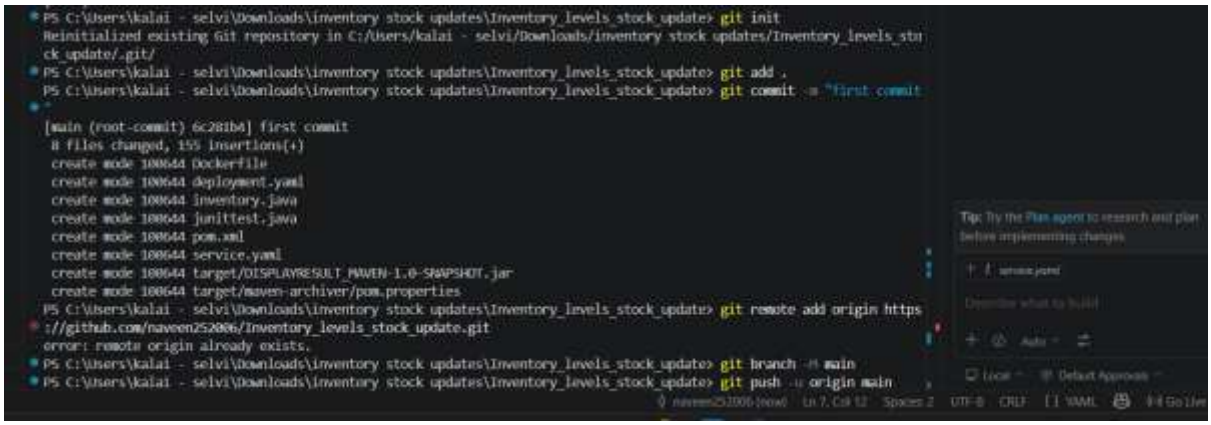
Git add .

git commit -m "first commit"

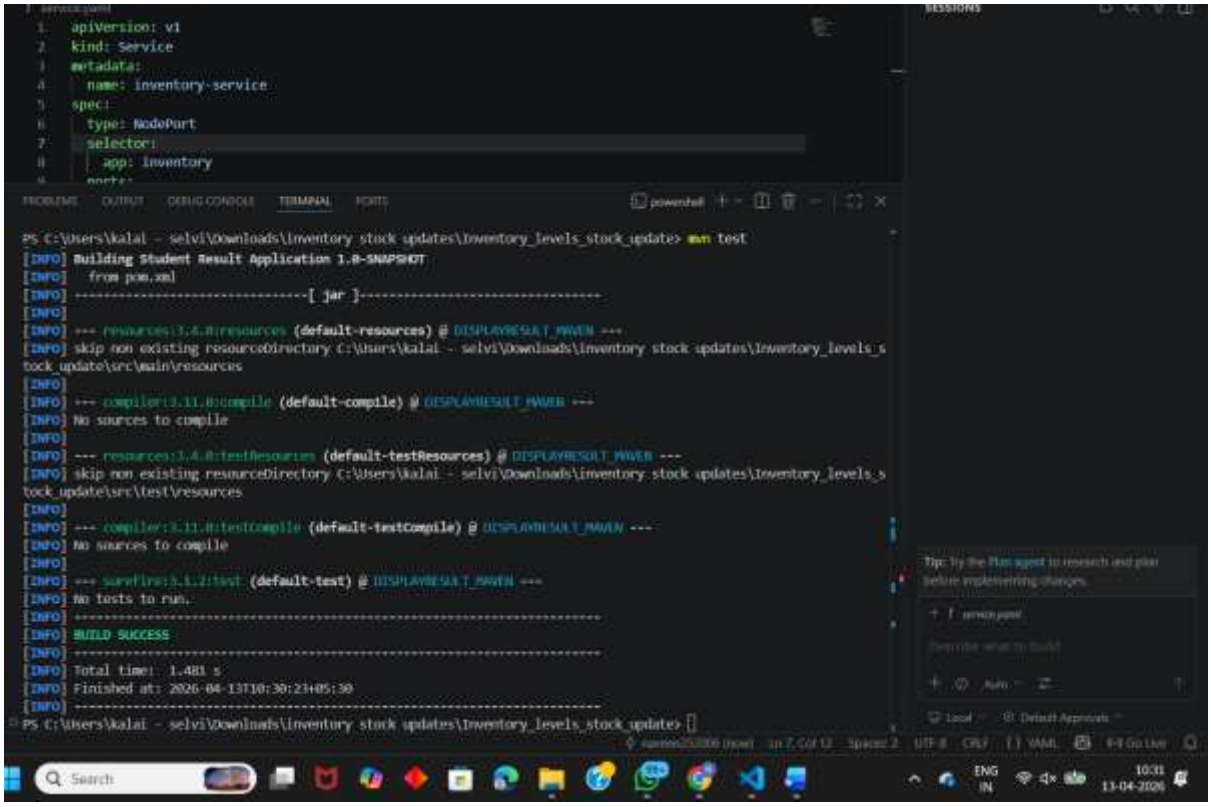
git branch -M main

git remote add origin https://github.com/naveen252006/Inventory_levels_stock_update.git

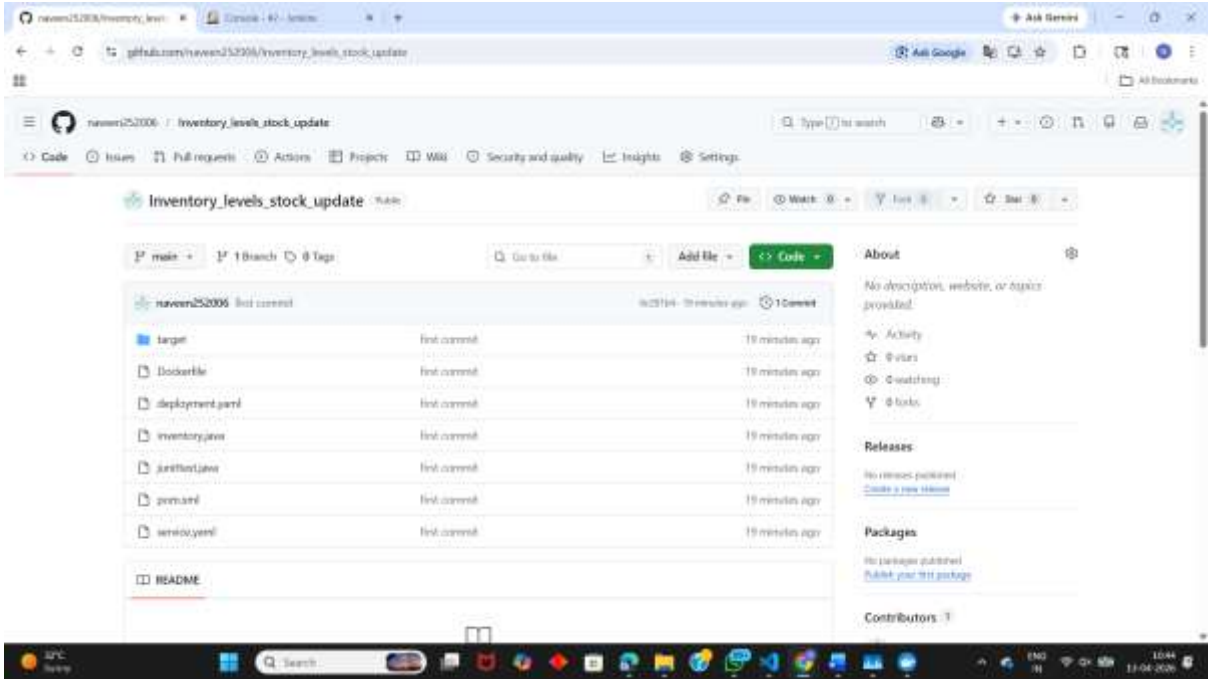
git push -u origin main



.



.



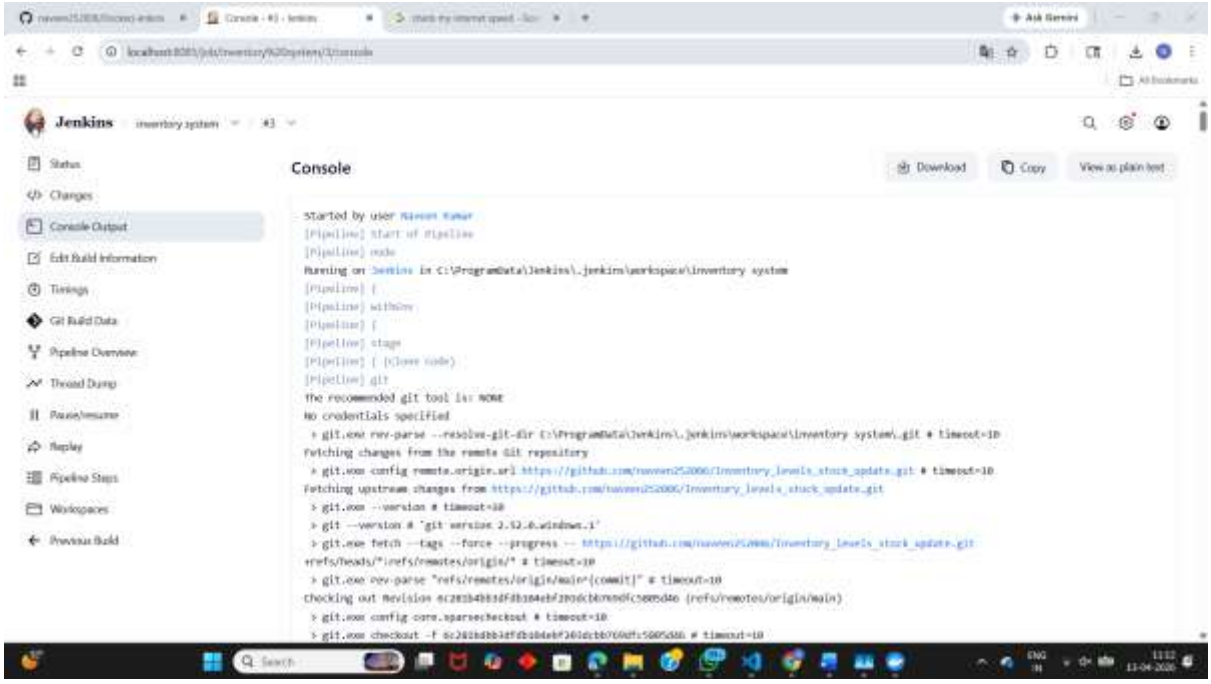
.

Jenkinspipeline:

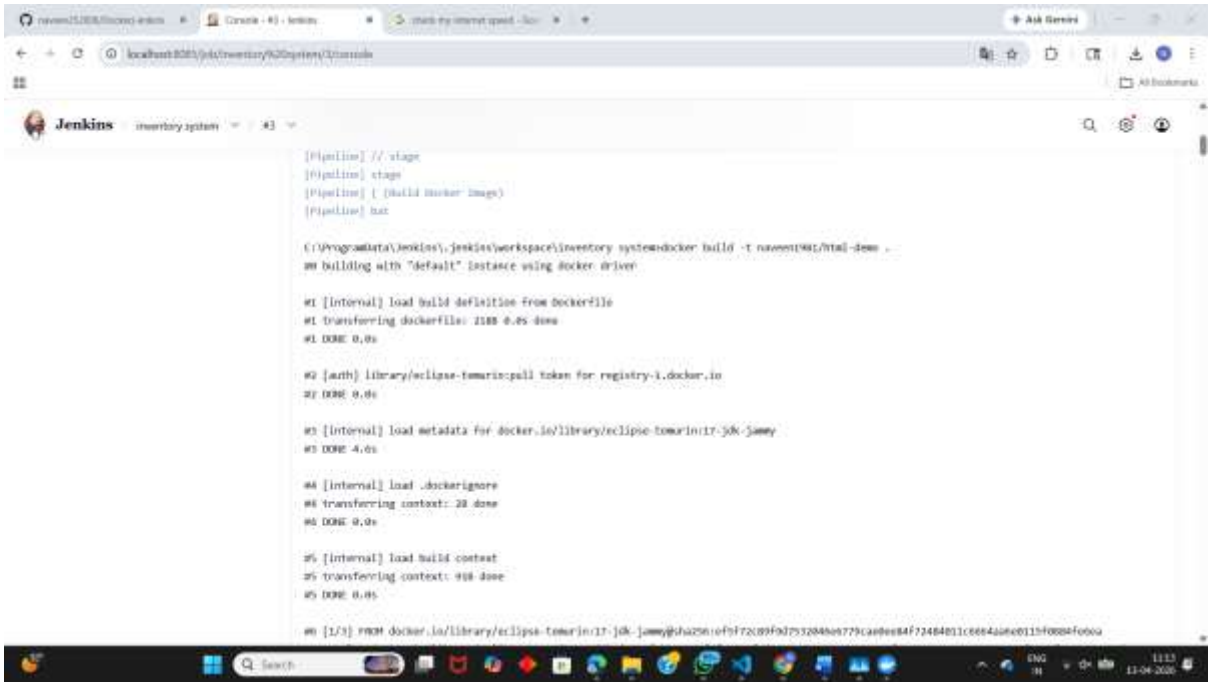
```
pipeline {
    agent any

    environment {
        DOCKER_IMAGE = "naveen1981/html-demo"
    }

    stages {

        stage('Clone Code') {
            steps {
                git branch: 'main',
```

```
            url:
'https://github.com/naveen252006/Inventory_levels_stock_update.git'
        }
    }

    stage('Build Docker Image') {
        steps {
            bat 'docker build -t %DOCKER_IMAGE% .'
        }
    }

    stage('Push Image') {
        steps {
            withCredentials([
                usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'USER',
                    passwordVariable: 'PASS'
                )
            ]) {
                bat 'docker login -u %USER% -p %PASS%'
                bat 'docker push %DOCKER_IMAGE%'
            }
        }
```

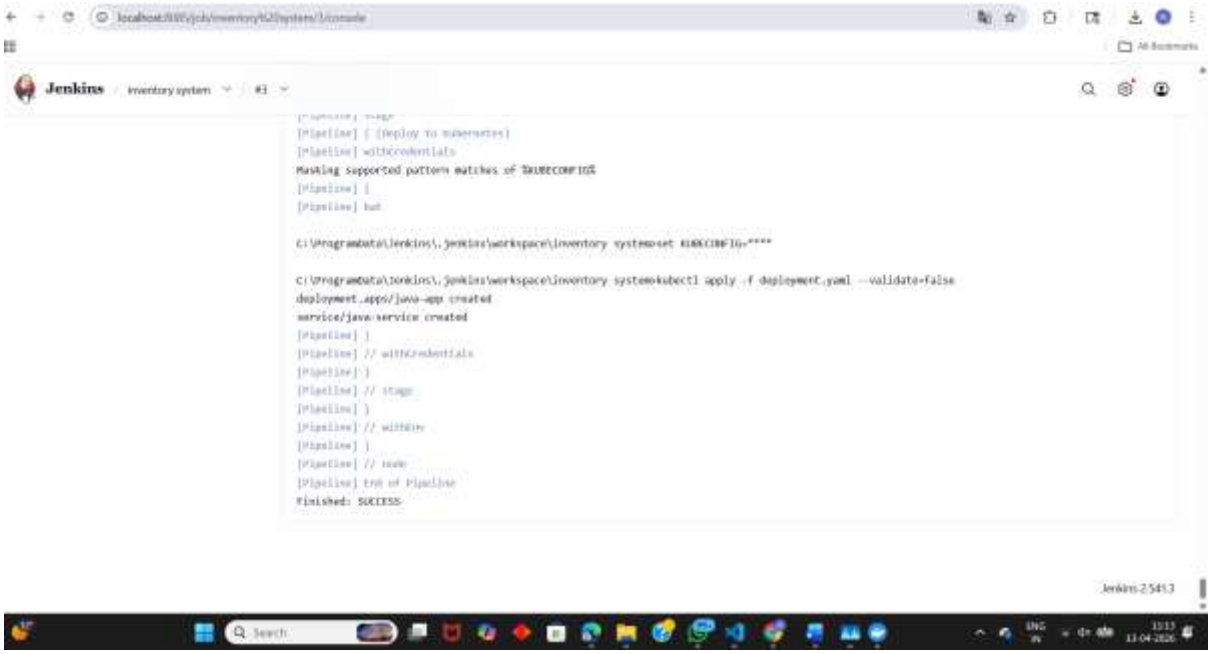
```
        }

        stage('Deploy to Kubernetes') {
            steps {
                withCredentials([s
                    file(
                        credentialsId: 'kubeconfig',
                        variable: 'KUBECONFIG'
                    )
                ]) {
                    bat '''
                    set KUBECONFIG=%KUBECONFIG%
                    kubectl apply -f deployment.yaml --validate=false
                    '''
                }
            }
        }

    }
}
```


Jenkins
inventory system
#3
Status
Changes
Console Output
Edit Build Information
Timings
Git Build Data
Pipeline Overview
Thread Dump
Pause/resume
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Naveen Kumar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\inventory system
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\inventory system\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/naveen252006/Inventory_levels_stock_update.git # timeout=10
Fetching upstream changes from https://github.com/naveen252006/Inventory_levels_stock_update.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/naveen252006/Inventory_levels_stock_update.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f

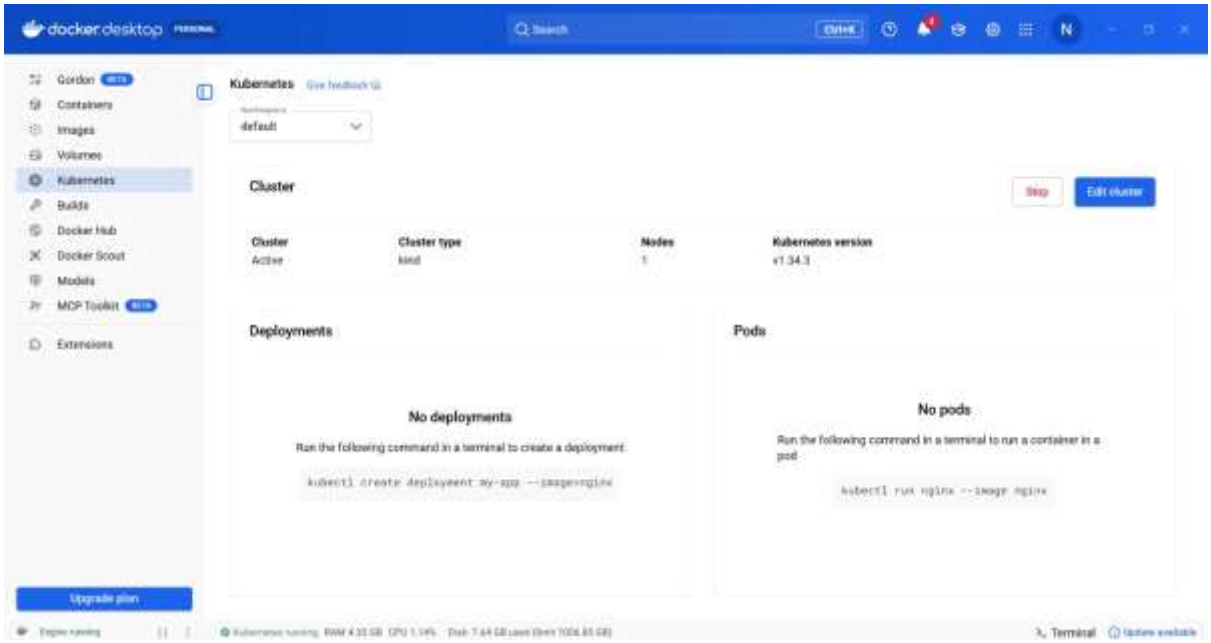
.


Jenkins
inventory system
#3
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\inventory system>docker build -t naveen981/html-demo .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Sending build context to Docker daemon 2.048kB
Step 1/2 : FROM eclipse-temurin:17-jdk-jammy
17-jdk-jammy: Pulling from library/eclipse-temurin
Digest: sha256:ef5f72c89f0d75320d4a6779ca0ee84f72484011c6664a6e0115f0884f06cc
Status: Downloaded newer image for eclipse-temurin:17-jdk-jammy
---> 4f2484011c6664
Step 2/2 : COPY index.html /usr/share/nginx/html/index.html
---> 3f0e0e0e0e0e
Successfully built 3f0e0e0e0e0e
Successfully tagged naveen981/html-demo:latest
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push Docker Image)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\inventory system>docker push naveen981/html-demo
The push refers to repository [docker.io/naveen981/html-demo]
latest: digest: sha256:3f0e0e0e0e0e size: 1234
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Container)
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\inventory system>docker run -d -p 8080:80 --name html-app naveen981/html-demo
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

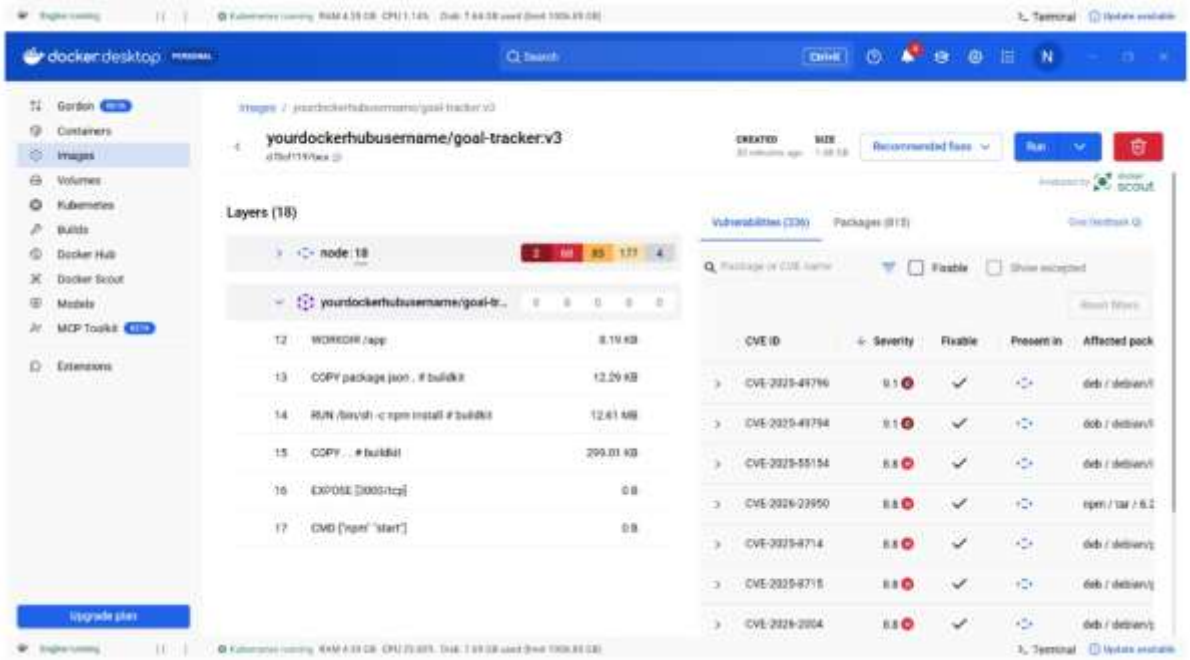
.

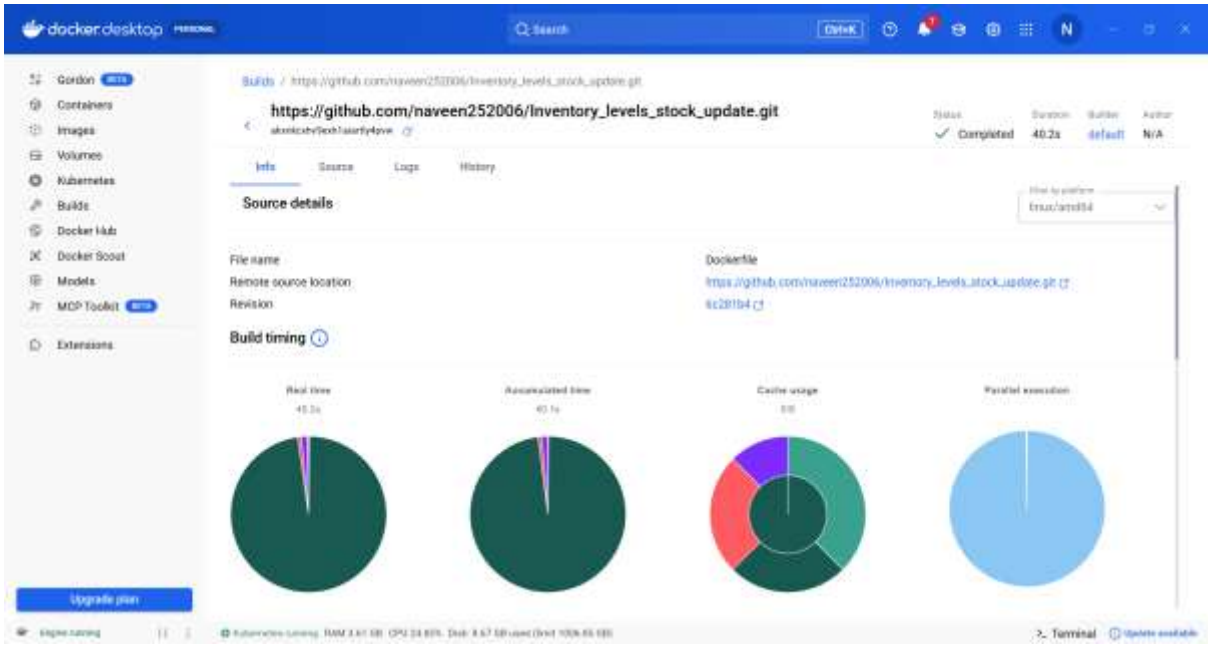

Jenkins
Masking supported pattern matches of $KUBECONFIG
deployment.apps/java-app created
service/java-service created
Finished: SUCCESS

Kubernets:


docker desktop
Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit
Extensions
Upgrade plan
Kubernetes
default
Cluster
Stop
Edit cluster
Cluster
Active
Cluster type
kind
Nodes
1
Kubernetes version
v1.34.3
Deployments
No deployments
Run the following command in a terminal to create a deployment
kubectl create deployment my-app --image=nginx
Pods
No pods
Run the following command in a terminal to run a container in a pod
kubectl run nginx --image nginx
Terminal

.


docker desktop
Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit
Extensions
Upgrade plan
yourdockerhubusername/goal-tracker:v3
Layers (18)
node:18
WORKDIR /app
COPY package.json . # buildkit
RUN /bin/sh -c npm install # buildkit
EXPOSE [3000/tcp]
CMD ["npm" "start"]
Run
Recommended fixes
Vulnerabilities (336)
Packages (813)
CVE ID
Severity
Fixable
Present in
Affected pack


docker desktop
Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit
Extensions
Upgrade plan
https://github.com/naveen252006/Inventory_levels_stock_update.git
Completed
40.2s
default
N/A
Info
Source
Logs
History
Source details
File name
Dockerfile
Remote source location
Revision
Build timing
Real time
Accumulated time
Cache usage
Parallel execution

.